

CL inflow fits with a virial window for the SPARC category-C galaxies: detection, model selection, fit quality and robustness

Automated analysis for E.P.J. de Haas

Summary

Category C contains the 41 SPARC galaxies whose single-Lagrangian (single-L) residuals show a significant $- + -$ wave: the data rise above the CL curve at intermediate radii and fall below it further out. For all of them we fitted the CL inflow model extended with a virial window, as introduced in de Haas (2026) [1] and in the CL field paper [2]. Beyond a radius r_v the orbital speed gains a Newtonian component of strength p . The window is accepted in 29 galaxies ($\Delta\text{BIC} < -6$ relative to single-L); in 12 the wave is too weak and single-L is kept. In 26 of the accepted galaxies the window is a *descent* ($p > 2$): beyond r_v , v^2 falls as GM_N/r onto a floor A . For these, median χ^2_ν falls from 4.28 to 1.01, RMS_{rel} from 0.117 to 0.073, and the window radius, floor and descent mass are determined to 9%, 6% and 23% (medians). The window is preferred over the two-Lagrangian alternative in every accepted galaxy, and a free gauge offset is never required. The descent galaxies are early and intermediate spirals (Sab–Sc), complementary to the late-type nested-spiral galaxies of category B. Residual structure persists in 18 of 26, and 7 remain at $\chi^2_\nu > 2$; these are bulge-dominated curves with very small errors, where the inner rise and peak need further regions.

1 Models

Single-L. The CL inflow curve of [2] (Eqs. 17–18; see also [1, Eqs. 4–5] and [3]), with $X = \sqrt{2GM/R} - H_z R$ and $Y(r) = \sqrt{2GM/r} - H_z r$:

$$v^2 = \frac{1}{2}X^2 r^2 / R^2 \quad (r \leq R), \quad v^2 = \frac{3}{2}X^2 - Y(r)^2 \quad (r > R), \quad H_z = 2.2 \times 10^{-18} \text{ s}^{-1}.$$

CL + virial window (VW). Following [1, Eq. 6] and [2, Eq. 19], beyond r_v :

$$v^2(r) = \frac{3}{2}X^2 - Y(r)^2 + p \left[\frac{1}{2}Y(r)^2 - \Phi_p \right], \quad r \geq r_v.$$

Two choices differ from [1], where r_v is set by inspection and Φ_p is free. Here r_v is fitted, and $\Phi_p = \frac{1}{2}Y(r_v)^2$ makes v^2 continuous at r_v ; leaving Φ_p free (model VWf, $k = 5$) is tested separately. For $H_z \rightarrow 0$ the window reads

$$v^2 = A + \frac{GM_N}{r}, \quad A = \frac{3GM}{R} - \frac{pGM}{r_v}, \quad M_N = (p - 2)M,$$

so $p > 2$ gives a Keplerian descent of mass M_N onto the floor A , $0 < p < 2$ a flattening, $p < 0$ an additional rise, and $p = 3$ the pure Kepler term of the inner mass M . Because p and M become degenerate when the descent dominates, the fit is parametrised by (M, R, r_v, M_N) , which are also the measured quantities. This descent behaviour was first noticed in earlier spreadsheet fits as solutions with formally negative (M, R) , which have exactly the $A + B/r$ form; the window replaces them with positive parameters valid for $H_z \neq 0$.

Alternatives. Two-L ([2, Eqs. 24–26]; $k = 4$, the same parameter count as VW), and VWf ($k = 5$).

Selection and uncertainties. The window is accepted when $\Delta\text{BIC} = \text{BIC}_{\text{VW}} - \text{BIC}_{\text{S}} < -6$, and assigned to two-L instead if two-L has lower BIC by more than 2. Uncertainties come from five sources. (1) Statistical: local covariance; A by linear propagation. (2) r_v : since $\partial v^2 / \partial r$ is discontinuous at r_v , $\chi^2(r_v)$ changes slope at each data radius, so a profile-likelihood interval ($\Delta\chi^2 \leq 1$, 80 grid values, other parameters refitted) is used. (3) Jackknife: leave-one-out. (4) Systematic: distance $+\delta D$ and inclination $+\delta i$ refits. (5) $H_z = 0$ refit. The profile scan also served as a global check of the optimum; it found a better solution for NGC 6674, which was adopted.

Grades (descent windows). **I:** R , r_v , M_N and A all within 30%, at least 3 points on each side of r_v , jackknife scatter $\leq 30\%$. **II:** A within 10% and M_N within 50%. **III:** otherwise. **M** flag: $\chi^2_\nu > 2$ after the window.

2 Model selection outcome

Table 1: Category-C outcome.

Outcome	N	Galaxies
Descent window, grade I	7	NGC2903, NGC3198, NGC3992, NGC4100, NGC5055, UGC02953, UGC05253
Descent window, grade II	7	NGC4157, NGC5907, NGC6503, NGC6674, NGC7331, UGC03205, UGC08699
Descent window, grade III	12	NGC0891, NGC3521, NGC4088, NGC4217, NGC5033, NGC5371, NGC5985, UGC02916, UGC05986, UGC11455, UGC11914, UGC12506
Flattening window ($0 < p < 2$)	2	ESO563-G021, IC4202
Rising window ($p < 0$)	1	NGC6946
Single-L kept	12	F568-V1, NGC1705, NGC2366, NGC3769, NGC4183, NGC4559, UGC01230, UGC04305, UGC05721, UGC06983, UGC08490, UGC09037

Galaxies with $\chi^2_\nu > 2$ after the window (flag M): NGC2903, NGC5055, NGC5371, NGC5907, NGC6674, NGC6946, UGC02953, UGC11455.

Window versus two-L. With the same number of parameters, the window beats the two-Lagrangian model in every accepted galaxy (smallest $\text{BIC}_{2\text{L}} - \text{BIC}_{\text{VW}} = +2.2$, median +43). A two-L reset produces an upward step and cannot follow a $1/r$ descent, so category C is genuinely different from category B. **Free gauge.** Leaving Φ_p free never helps ($\Delta\text{BIC}_{\text{VWf-VW}} \geq -1.0$), so the data show no step at r_v and the continuous gauge is adequate. **Single-L kept.** The 12 rejected galaxies already have small single-L χ^2_ν (median 0.48) and are mostly late types; their $- + -$ wave lies within the SPARC errors. **Other windows.** ESO563-G021 and IC 4202 need an outer *flattening* ($p \approx 1.7$), and NGC 6946 needs an extra *rise* ($p < 0$, with two-L only +2.2 behind). That makes NGC 6946 a borderline case between categories B and C.

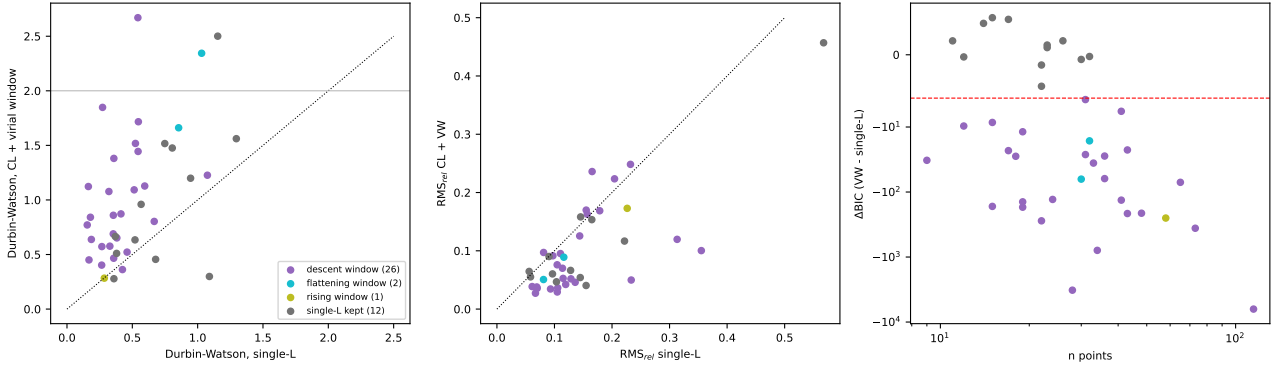


Figure 1: Left: Durbin–Watson statistic before and after the window. Middle: relative RMS in v^2 . Right: $\Delta\text{BIC (VW - single-L)}$ versus n ; acceptance threshold -6 . Colours: descent (purple), flattening (cyan), rising (olive), single-L kept (grey).

3 Fit quality of the descent windows

The window removes the dominant $- + -$ pattern in all descent galaxies. The median Durbin–Watson statistic rises from 0.36 to 0.85, and the residual sign pattern loses its long runs (atlas). Residual structure is not fully removed in 18 galaxies. Unlike categories A and B, category C contains many high-precision, bulge-dominated curves (median $n = 31$, 20 of 26 $Q=1$). There small, coherent deviations near the inner peak become significant: 10 galaxies have at least one residual beyond 3σ , and Shapiro–Wilk rejects normality in 9. The SPARC errors are again conservative on average (5 galaxies with $P(\chi^2 \leq \chi^2_{\text{obs}}) < 0.01$), so $\chi^2_\nu < 1$ is not unusual. The flag-M galaxies (NGC2903, NGC5055, NGC5371, NGC5907, NGC6674, NGC6946, UGC02953, UGC11455) are improved by factors of 2 to 30 in χ^2 , but their inner bulge region and peak are not described by a single CL bulge. They are the natural targets for a bulge Lagrangian plus disk Lagrangian with a window, or for the two-window scheme of [1, Eqs. 5–7].

Table 2: Descent-window galaxies: fit-quality statistics (medians, single-L $\rightarrow$ CL + virial window).

	Grade I	Grade II	Grade III
χ^2_ν	6.72 $\rightarrow$ 0.75	4.58 $\rightarrow$ 1.03	2.21 $\rightarrow$ 1.02
RMS _{rel}	0.232 $\rightarrow$ 0.100	0.129 $\rightarrow$ 0.052	0.108 $\rightarrow$ 0.084
Durbin–Watson	0.33 $\rightarrow$ 0.77	0.36 $\rightarrow$ 0.87	0.37 $\rightarrow$ 0.94
$p_{\text{runs}} < 0.05$ after window	6/7	5/7	7/12
ΔBIC (median)	-361	-133	-25
n (median)	34	31	26

4 Parameter determination

Median relative uncertainties for the descent windows are: R 14%, r_v 9% (profile), M_N 23%, A 5.8%, and M 16%. The floor A and the window radius are the best-determined quantities. The parametrisation by M_N removes the p – M degeneracy (median $\rho(M, M_N) = 0.00$). The virial strength $p = 2 + M_N/M$ is then a derived quantity, and it becomes very large where M is small: in NGC 891, NGC 5033, NGC 6674, UGC 2916 and UGC 5253 the inner CL bulge shrinks and the curve is a compact central mass plus Keplerian descent. p ranges over 3.8–61 (16–84%, median 7.1), and 8 galaxies have $p \leq 5$. The pure Kepler case $p = 3$ is therefore one end of a broad distribution, not a universal value.

The r_v likelihood. Fig. 2 shows that $\chi^2(r_v)$ has a well-defined minimum in most galaxies, kinked at the data radii. Because v^2 is continuous at r_v , the profile is smoother than the R_2 profile of the two-L model, and r_v is localised to within one or two point spacings.

Systematics, jackknife, H_z . Distance and inclination systematics are comparable to or larger than the statistical errors (median sys/stat 1.7 for r_v , 0.9 for A , 0.5 for M_N). Leave-one-out scatter stays below 30% for grade I. Setting $H_z = 0$ changes the parameters by at most about 12% except in NGC 5033 and UGC 2916, where the inner R is poorly constrained and the solution moves along the M – R degeneracy.

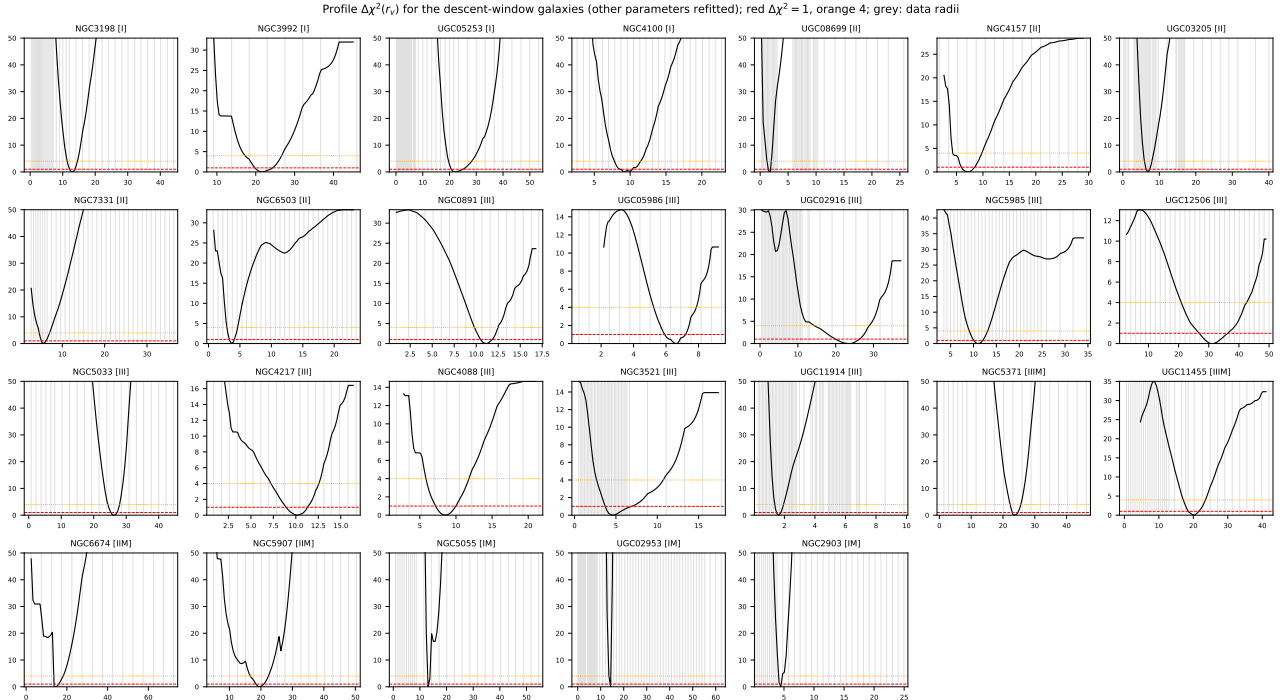


Figure 2: Profile $\Delta\chi^2(r_v)$ for the descent-window galaxies; grey lines are the data radii.

5 Physical consistency

- **Window radius.** $r_v = 4.6$ –21.7 kpc (median 11.1), $r_v/R_{\text{disk}} = 3.4$ (16–84%: 1.8–4.2; Spearman $\rho = 0.76$). The descent sets in at 2–4 disk scale lengths, beyond the rotation-curve peak.

- **Floor.** $\sqrt{A}$ tracks SPARC V_{flat} ($r = 0.92$), with $\sqrt{A}/V_{\text{flat}} = 0.92$. The floor is the asymptotic speed that the descent approaches; V_{flat} , measured over the last points, lies slightly above it because the descent is not yet complete.
- **Descent mass.** $M_N/M_{\text{bar}} = 0.58$ (16–84%: 0.14–1.42; $r = 0.63$ in log). For the 13 galaxies with a photometric bulge, M_N is 2.8 times the bulge stellar mass ($\Upsilon_{\text{bulge}} = 0.7$; 16–84%: 0.8–8.8). The Keplerian descent mass is therefore of the order of the baryonic mass inside the peak, consistent with the reading in [1] that the window matter orbits Newtonially with respect to the inflowing metric.
- **Morphology.** The descent windows are early and intermediate spirals (Sab 6, Sb 6, Sbc 7, Sc 3, Scd 3, Sm 1), with 20 of 26 Q=1. The single-L-kept galaxies are mostly Scd–BCD. Together with category B (late-type nested spirals), this gives a clean morphological split. Bulge-dominated galaxies show a Keplerian descent window; bulgeless late types show a Lagrangian reset.

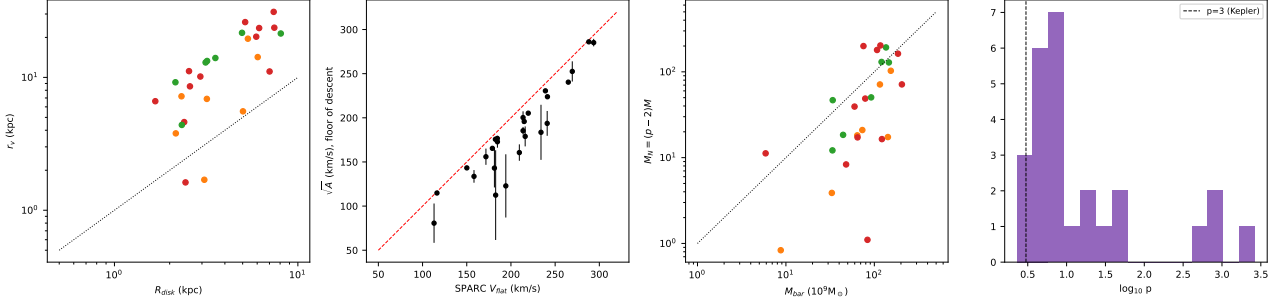


Figure 3: Descent windows. Left to right: r_v versus R_{disk} ; floor $\sqrt{A}$ versus SPARC V_{flat} ; descent mass M_N versus M_{bar} ; distribution of the virial strength p (dashed: pure Kepler $p = 3$). Colours: grade I green, II orange, III red.

6 Assessment

The descent is a robust, recurring feature. The CL + virial window of [1,2], with r_v fitted and a continuous gauge, describes 26 of the 41 category-C galaxies far better than single-L (ΔBIC from -6 to -6318). It is always preferred over a two-Lagrangian reset and needs no free gauge. Its measured quantities, the window radius r_v , the floor A and the Keplerian descent mass M_N , are well determined and match independent photometric and kinematic scales.

Grades. Grade I (NGC2903, NGC3198, NGC3992, NGC4100, NGC5055, UGC02953, UGC05253) is the robust core. Grade II (NGC4157, NGC5907, NGC6503, NGC6674, NGC7331, UGC03205, UGC08699) has a well-determined floor and window but a less certain descent mass. Grade III (NGC0891, NGC3521, NGC4088, NGC4217, NGC5033, NGC5371, NGC5985, UGC02916, UGC05986, UGC11455, UGC11914, UGC12506) has a weakly determined descent mass ($> 50\%$) or floor ($> 10\%$), mostly because only a few points lie in the window, the descent is shallow (p close to 2), or the inner radius is unconstrained.

Limits. (i) The inner bulge region of the most massive, best-measured spirals is not captured by a single CL bulge, which leaves residual structure in about two thirds of the descent galaxies and $\chi^2_\nu > 2$ in eight. (ii) p is not a robust quantity when the inner mass is small; M_N and A should be reported instead. (iii) The window has no outer end here. The two-window scheme of [1] (window strengths p, q) is the natural next refinement for the flag-M galaxies, together with an explicit bulge component.

References

- [1] E.P.J. de Haas (2026), From Rotation Curves to Cosmic Time: Probing High-Redshift Expansion through Nested Spiral Dynamics, *J. High Energy Phys. Grav. Cosmol.* 12, 334–367, doi:10.4236/jhepgc.2026.121022 (virial term Eq. 3; multi-region virial model Eqs. 5–7).
- [2] E.P.J. de Haas, Galactic Rotation Curves and the Constant–Lagrangian Field: Empirical Tests within the Q_g Rotor Framework (manuscript; single-L Eqs. 17–18, virial term Eq. 19, two-L Eqs. 24–26).
- [3] E.P.J. de Haas, The Role of the Hubble Parameter in Galactic Rotation Curves and Spiral Morphology (manuscript).
- [4] F. Lelli, S.S. McGaugh, J.M. Schombert (2016), SPARC, *AJ* 152, 157.

Table 3: All category-C galaxies: single-L, CL + virial window (VW), two-L and VWf comparison.

Galaxy	Type	Q	n	$\chi^2_{L,S}$	RMS _S	$p_{\text{runs},S}$	$\chi^2_{L,VW}$	ΔBIC_{VW}	$\text{BIC}_{2L}-\text{BIC}_{VW}$	$\text{BIC}_{VWf}-\text{BIC}_{VW}$	p	Outcome	Grade
UGC02953	Sab	2	115	68.64	0.232	1.2e-24	12.87	-6317.8	+5081.6	+4.7	35.8	descent window	IM
NGC5055	Sbc	1	28	128.68	0.313	2.9e-06	4.36	-3234.6	+732.1	+3.3	14.5	descent window	IM
NGC2903	Sbc	1	34	26.86	0.355	1.0e-07	2.10	-789.5	+543.6	+3.5	5.19	descent window	IM
UGC05253	Sab	2	73	5.54	0.093	3.0e-15	0.35	-360.7	+10.8	+1.1	496	descent window	I
NGC5033	Sc	1	22	15.11	0.119	5.3e-05	1.04	-277.3	+84.2	-1.0	2.68e+03	descent window	III
NGC6946	Scd	1	58	9.71	0.226	8.4e-09	5.27	-251.4	+2.2	+4.1	-284	rising window	-
NGC3198	Sc	1	43	6.13	0.204	5.7e-09	0.75	-214.2	+219.2	+3.8	3.8	descent window	I
UGC03205	Sab	1	48	5.93	0.179	1.2e-07	1.21	-211.8	+218.1	+3.4	6.36	descent window	II
NGC5371	Sbc	1	19	14.64	0.110	2.5e-04	4.79	-171.2	+89.3	+2.9	40.2	descent window	IIIM
NGC6674	Sb	1	15	16.38	0.165	3.1e-02	3.74	-166.4	+172.2	+21.2	818	descent window	IIIM
NGC5907	Sc	1	19	10.88	0.070	2.1e-04	2.50	-141.6	+147.9	+2.9	15.2	descent window	IIIM
UGC08699	Sab	2	41	4.58	0.114	1.2e-07	1.03	-133.3	+98.5	+3.7	7.8	descent window	II
NGC4100	Sbc	1	24	6.72	0.233	1.6e-05	0.57	-130.0	+64.5	+3.2	6.28	descent window	I
UGC11914	Sab	1	65	2.22	0.081	9.0e-14	0.99	-70.9	+79.3	+4.2	2.27	descent window	III
ESO563-G021	Sbc	1	30	3.38	0.116	4.2e-06	0.94	-63.3	+29.0	+3.4	1.76	flattening window	-
NGC7331	Sb	1	36	2.70	0.067	4.0e-04	0.71	-62.2	+70.2	+3.6	3.96	descent window	II
NGC5985	Sb	1	33	2.20	0.069	1.4e-03	0.88	-35.9	+42.9	+3.0	2.63	descent window	III
NGC3992	Sbc	1	9	5.62	0.105	3.9e-02	0.48	-32.5	+22.8	+2.2	9.26	descent window	I
NGC0891	Sb	1	18	3.06	0.097	1.9e-05	1.06	-28.2	+54.4	+2.5	61.5	descent window	III
UGC11455	Scd	1	36	3.97	0.155	2.0e-07	3.12	-28.0	+21.2	+3.5	4.15	descent window	IIIM
NGC6503	Scd	1	31	1.92	0.136	3.9e-03	0.82	-26.5	+33.4	-0.8	2.78	descent window	II
NGC4157	Sb	1	17	2.10	0.129	9.9e-04	0.21	-23.1	+26.6	+2.8	3.68	descent window	II
UGC02916	Sab	2	43	2.48	0.115	1.1e-06	1.84	-22.5	+19.0	+3.8	848	descent window	III
IC4202	Sbc	1	32	1.31	0.081	2.1e-02	0.57	-16.3	+10.9	+3.5	1.65	flattening window	-
NGC4217	Sb	1	19	1.11	0.106	2.1e-04	0.08	-11.8	+8.8	+2.9	7.77	descent window	III
NGC4088	Sbc	1	12	1.81	0.105	8.3e-03	0.40	-9.9	+10.0	+2.5	3.94	descent window	III
UGC05986	Sm	2	15	2.15	0.144	8.1e-03	1.20	-9.4	+7.3	+2.7	5.84	descent window	III
NGC3521	Sbc	1	41	0.45	0.061	2.9e-06	0.07	-7.8	+10.5	+3.7	6.03	descent window	III
UGC12506	Scd	2	31	0.71	0.157	2.6e-06	0.28	-6.2	+6.0	+3.4	16.7	descent window	III
UGC09037	Scd	2	22	0.86	0.091	2.7e-04	0.37	-4.4	+1.6	+2.9	-2.42	single-L kept	-
UGC04305	Im	3	22	0.65	0.568	1.3e-03	0.30	-1.4	+4.2	+3.1	2.2	single-L kept	-
UGC08490	Sm	1	30	0.50	0.103	1.3e-04	0.25	-0.6	-0.1	+3.4	-4.95	single-L kept	-
NGC3769	Sb	2	12	0.79	0.097	1.1e-02	0.32	-0.3	+4.0	+2.3	4.79	single-L kept	-
NGC4559	Scd	1	32	0.44	0.155	1.6e-06	0.21	-0.2	+0.0	+3.5	-12.7	single-L kept	-
UGC05721	Sd	1	23	1.01	0.222	1.4e-04	0.84	1.0	-0.1	+3.1	-12.3	single-L kept	-
NGC4183	Scd	1	23	0.54	0.145	6.9e-04	0.34	1.3	-0.1	+3.1	-8.64	single-L kept	-
UGC01230	Sm	1	11	0.37	0.128	1.3e-02	0.07	1.9	+0.8	+2.4	2.14	single-L kept	-
NGC2366	Im	3	26	0.47	0.165	3.2e-05	0.30	1.9	+2.5	+3.3	4.11	single-L kept	-
NGC1705	BCD	3	14	0.15	0.058	4.9e-03	0.08	4.3	+0.9	+2.0	4.27	single-L kept	-
UGC06983	Scd	1	17	0.32	0.056	1.4e-02	0.31	4.9	-0.3	+2.8	-1.13	single-L kept	-
F568-V1	Sd	1	15	0.08	0.145	8.1e-03	0.07	5.1	+0.1	+2.7	1.11	single-L kept	-

Table 4: Accepted windows: parameters. R, r_v in kpc; M, M_N in $10^9 M_\odot$; A in $\text{km}^2 \text{s}^{-2}$. R, M_N, A : statistical $\pm$ systematic; r_v : profile $\Delta\chi^2 \leq 1$ interval [systematic]. $n_{\text{in}}/n_{\text{win}}$: points before/after r_v .

Galaxy	Grade	n	R	M	r_v	M_N	A	p	$n_{\text{in}}/n_{\text{win}}$	jk
UGC05253	I	73	0.0561 $\pm$ 0.015 $\pm$ 0.013	0.264 $\pm$ 0.074	21.4 $^{+3.1}_{-0.17}$ [5.4]	130 $\pm$ 11 $\pm$ 41	34364 $\pm$ 1.6e + 03 $\pm$ 5.6e + 03	496	62/11	0.20
NGC3198	I	43	3 $\pm$ 0.083 $\pm$ 0.3	6.74 $\pm$ 0.27	13 $^{+0.75}_{-0.78}$ [1.3]	12.2 $\pm$ 2.6 $\pm$ 1.2	20525 $\pm$ 4.6e + 02 $\pm$ 5.9e + 02	3.8	27/16	0.19
NGC4100	I	24	2.82 $\pm$ 0.22 $\pm$ 0.39	10.9 $\pm$ 1.3	9.19 $^{+1.1}_{-1.43}$ [1.3]	46.7 $\pm$ 5.9 $\pm$ 6.5	17860 $\pm$ 1.9e + 03 $\pm$ 3.5e + 02	6.28	8/16	0.16
NGC3992	I	9	3.96 $\pm$ 1.2 $\pm$ 0.38	26.5 $\pm$ 9.9	21.7 $^{+2.5}_{-1.6}$ [2.1]	193 $\pm$ 40 $\pm$ 20	37508 $\pm$ 5.5e + 03 $\pm$ 1.6e + 03	9.26	3/6	0.09

Galaxy	Grade	n	R	M	r_v	M_N	A	p	n_{in}/n_{win}	jk
UGC03205	II	48	$1.01 \pm 0.074 \pm 0.2$	4.81 ± 0.44	$6.89^{+0.42}_{-0.06}$ [1.4]	$21 \pm 2.5 \pm 4.3$	$42131 \pm 6e + 02 \pm 2.2e + 03$	6.36	19/29	0.35
UGC08699	II	41	$0.196 \pm 0.082 \pm 0.049$	0.669 ± 0.31	$1.7^{+0.1}_{-0.1}$ [0.42]	$3.88 \pm 0.36 \pm 0.99$	$30856 \pm 4.2e + 02 \pm 2.1e + 03$	7.8	7/34	0.48
NGC7331	II	36	$1.42 \pm 0.5 \pm 0.15$	8.87 ± 3.9	$5.55^{+0.45}_{-0.38}$ [0.56]	$17.4 \pm 2.7 \pm 1.8$	$53150 \pm 6.9e + 02 \pm 2.2e + 02$	3.96	6/30	0.13
NGC6503	II	31	$0.829 \pm 0.13 \pm 0.041$	1.06 ± 0.21	$3.79^{+0.14}_{-0.42}$ [0.19]	$0.833 \pm 0.41 \pm 0.044$	$13182 \pm 1.7e + 02 \pm 2.4e + 02$	2.78	4/27	0.42
NGC4157	II	17	$2.61 \pm 0.38 \pm 0.36$	10.8 ± 2.5	$7.21^{+1.2}_{-1.2}$ [1]	$18.2 \pm 8 \pm 2.5$	$29881 \pm 2.3e + 03 \pm 3.5e + 02$	3.68	4/13	0.18
NGC5033	III	22	$0.02 \pm 0.018 \pm 0$	0.0745 ± 0.066	$26.1^{+0.78}_{-0.31}$ [10]	$200 \pm 58 \pm 3.7e + 02$	$15096 \pm 8.8e + 03 \pm 3.5e + 04$	$2.68e+03$	13/9	0.30
UGC11914	III	65	$0.489 \pm 0.028 \pm 0.15$	4.03 ± 0.34	$1.62^{+0.15}_{-0.085}$ [0.49]	$1.1 \pm 0.79 \pm 0.42$	$81826 \pm 7.6e + 02 \pm 1.9e + 04$	2.27	15/50	1.25
NGC5985	III	33	$3.13 \pm 0.17 \pm 0.78$	26.2 ± 2.2	$11.1^{+0.98}_{-0.61}$ [2.8]	$16.4 \pm 11 \pm 4.2$	$81325 \pm 2.4e + 03 \pm 3.1e + 03$	2.63	10/23	0.51
NGC0891	III	18	$0.206 \pm 0.14 \pm 0.01$	0.82 ± 0.56	$11.1^{+0.61}_{-0.59}$ [0.56]	$48.8 \pm 13 \pm 2.4$	$32016 \pm 4e + 03 \pm 29$	61.5	11/7	3.05
UGC02916	III	43	$0.0602 \pm 0.04 \pm 0.0087$	0.213 ± 0.14	$23.5^{+2.5}_{-2.5}$ [3.6]	$180 \pm 84 \pm 36$	$12638 \pm 1.1e + 04 \pm 1.7e + 03$	848	38/5	1.73
NGC4217	III	19	$2.05 \pm 0.14 \pm 0.28$	6.83 ± 0.67	$10.1^{+0.98}_{-1.2}$ [1.4]	$39.4 \pm 18 \pm 5.4$	$20464 \pm 6.3e + 03 \pm 78$	7.77	11/8	0.10
NGC4088	III	12	$2.73 \pm 0.43 \pm 0.38$	8.87 ± 2.2	$8.56^{+1.4}_{-1.2}$ [1.2]	$17.2 \pm 8.7 \pm 2.4$	$24314 \pm 2.9e + 03 \pm 6.1e + 02$	3.94	4/8	0.40
UGC05986	III	15	$2.14 \pm 0.078 \pm 0.64$	2.92 ± 0.18	$6.62^{+0.56}_{-2.1}$ [2]	$11.2 \pm 6.3 \pm 3.3$	$6506 \pm 3.6e + 03 \pm 17$	5.84	10/5	0.20
NGC3521	III	41	$0.516 \pm 0.053 \pm 0.15$	2.07 ± 0.26	$4.61^{+2.1}_{-0.89}$ [1.4]	$8.33 \pm 5.7 \pm 2.5$	$40114 \pm 2.8e + 03 \pm 1.5e + 03$	6.03	24/17	0.08
UGC12506	III	31	$2.71 \pm 0.39 \pm 0.27$	13.8 ± 2.3	$31.1^{+5}_{-3.8}$ [3.1]	$203 \pm 1.1e + 02 \pm 20$	$33696 \pm 1.1e + 04 \pm 1.9e + 02$	16.7	19/12	0.16
NGC5371	IIIM	19	$0.965 \pm 0.13 \pm 0.24$	4.26 ± 0.62	$23.7^{+0.56}_{-0.56}$ [5.9]	$163 \pm 23 \pm 39$	$25802 \pm 3e + 03 \pm 1.3e + 03$	40.2	11/8	0.64
UGC11455	IIIM	36	$4.6 \pm 0.15 \pm 0.69$	33.2 ± 1.8	$20.2^{+1}_{-1.3}$ [3]	$71.4 \pm 42 \pm 11$	$63773 \pm 5.8e + 03 \pm 0.2$	4.15	25/11	0.55
NGC6674	IIM	15	$0.02 \pm 0.0066 \pm 3.5e - 16$	0.126 ± 0.042	$14.2^{+1.3}_{-1.3}$ [5.3]	$103 \pm 13 \pm 61$	$50114 \pm 8.9e + 02 \pm 8.8e + 03$	818	3/12	0.18
NGC5907	IIM	19	$1.24 \pm 0.42 \pm 0.064$	5.42 ± 2	$19.6^{+1.3}_{-1.1}$ [1]	$71.3 \pm 8.5 \pm 3.6$	$38363 \pm 1.2e + 03 \pm 53$	15.2	6/13	0.55
UGC02953	IM	115	$0.493 \pm 0.0098 \pm 0.15$	3.81 ± 0.081	$14^{+0.32}_{-0}$ [4.2]	$129 \pm 2.7 \pm 39$	$57787 \pm 5.1e + 02 \pm 5.9e + 03$	35.8	77/38	0.13
NGC5055	IM	28	$1.12 \pm 0.059 \pm 0.034$	4.02 ± 0.24	$13.3^{+0.23}_{-0}$ [0.4]	$50.4 \pm 1.1 \pm 6.5$	$27360 \pm 2.2e + 02 \pm 3.3e + 03$	14.5	13/15	0.24
NGC2903	IM	34	$1.23 \pm 0.058 \pm 0.37$	5.78 ± 0.4	$4.39^{+0.689}_{-0}$ [1.3]	$18.4 \pm 0.98 \pm 5.5$	$31210 \pm 4.5e + 02 \pm 1.3e + 03$	5.19	12/22	0.12
ESO563-G021	flattening	30	$5.81 \pm 0.15 \pm 0.87$	65.8 ± 4	$11.3^{+1.8}_{-0.92}$ [1.7]	$-16 \pm 14 \pm 2.4$	$102140 \pm 3.1e + 03 \pm 1e + 03$	1.76	11/19	0.49
IC4202	flattening	32	$3.17 \pm 0.075 \pm 0.32$	18.5 ± 0.81	$9.45^{+0.63}_{-0.63}$ [0.94]	$-6.56 \pm 9.5 \pm 0.63$	$61324 \pm 2.8e + 03 \pm 9.4$	1.65	12/20	0.77
NGC6946	rising	58	$0.02 \pm 0.045 \pm 1.4e - 11$	0.0271 ± 0.062	$2.02^{+0}_{-0.077}$ [0.59]	$-7.76 \pm 0.51 \pm 2.3$	$33871 \pm 6.3e + 02 \pm 2.8e + 03$	-284	6/52	6.75

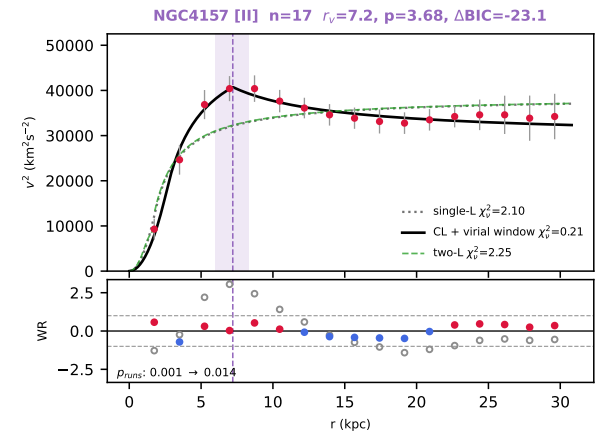
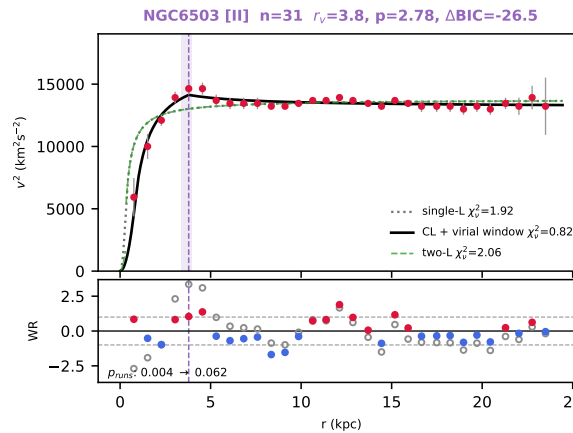
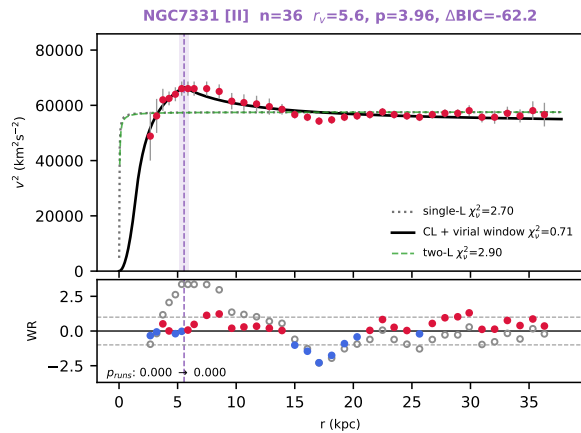
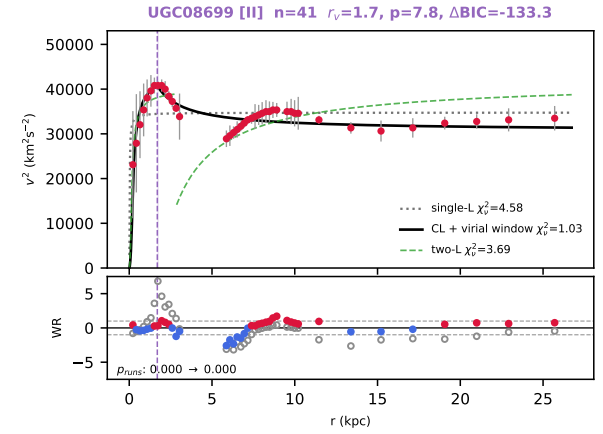
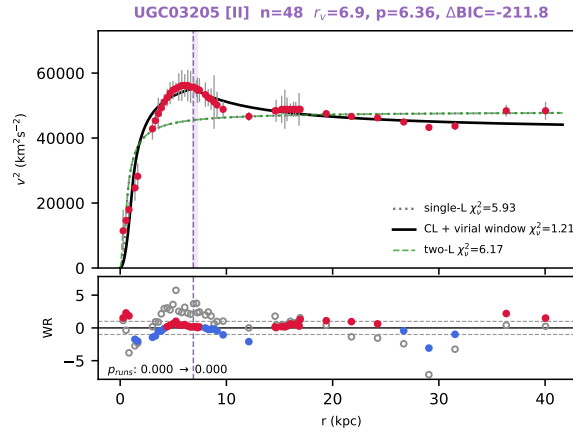
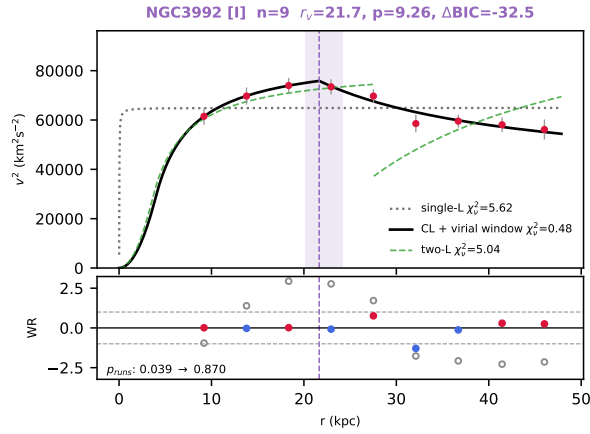
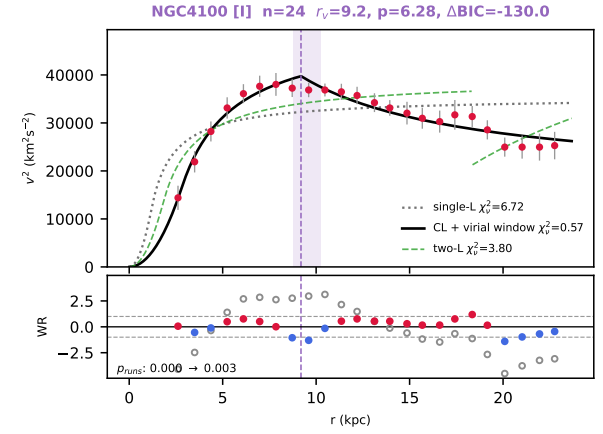
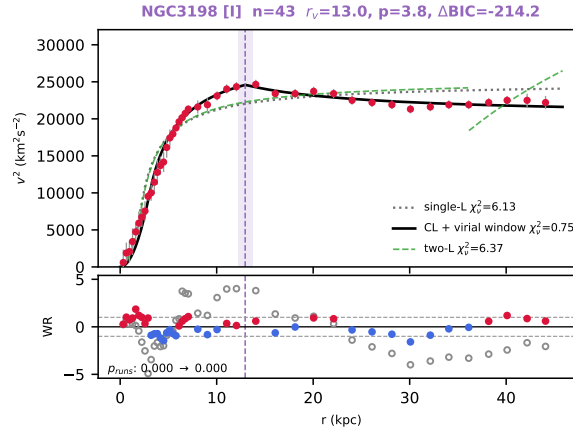
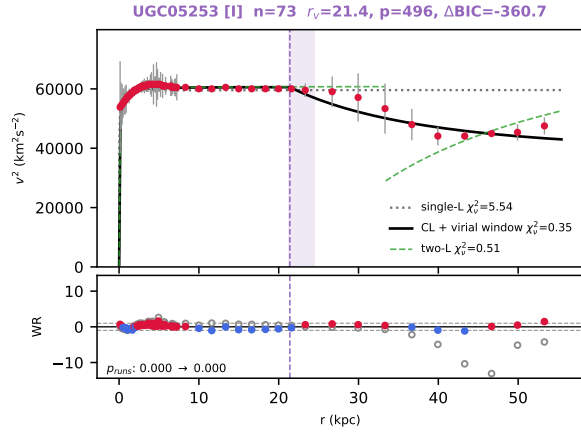
Table 5: Accepted windows: goodness of fit. $P_{<} = P(\chi^2 \leq \chi_{\text{obs}}^2)$ for $n - 4$ dof; p_{sw} Shapiro–Wilk; $\max|z|$ largest weighted residual.

Galaxy	$\chi^2_{\nu,S}$	$\chi^2_{\nu,W}$	BIC _S	BIC _{VW}	RMS _S	RMS _{VW}	$p_{\text{runs,VW}}$	DW _S $\rightarrow$ DW _{VW}	p_{sw}	$P_{<}$	$\max z $
UGC05253	5.54	0.35	402.1	41.4	0.093	0.034	0.000	0.33 $\rightarrow$ 0.58	0.159	0.000	1.62
NGC3198	6.13	0.75	258.6	44.4	0.204	0.224	0.000	0.19 $\rightarrow$ 0.64	0.177	0.132	1.86
NGC4100	6.72	0.57	154.2	24.2	0.233	0.050	0.003	0.15 $\rightarrow$ 0.77	0.179	0.067	1.41
NGC3992	5.62	0.48	43.7	11.2	0.105	0.029	0.870	0.54 $\rightarrow$ 2.67	0.043	0.210	1.29
UGC03205	5.93	1.21	280.7	68.9	0.179	0.169	0.000	0.41 $\rightarrow$ 0.87	0.046	0.843	3.08
UGC08699	4.58	1.03	186.1	52.8	0.114	0.070	0.000	0.27 $\rightarrow$ 0.40	0.013	0.576	2.58
NGC7331	2.70	0.71	99.1	36.9	0.067	0.027	0.000	0.17 $\rightarrow$ 0.45	0.007	0.110	2.29
NGC6503	1.92	0.82	62.5	35.9	0.136	0.046	0.062	0.59 $\rightarrow$ 1.13	0.525	0.273	1.89
NGC4157	2.10	0.21	37.1	14.1	0.129	0.052	0.014	0.36 $\rightarrow$ 1.38	0.144	0.001	0.71
NGC5033	15.11	1.04	308.4	31.1	0.119	0.042	0.042	0.16 $\rightarrow$ 1.12	0.715	0.593	2.24
UGC11914	2.22	0.99	148.2	77.3	0.081	0.097	0.000	0.67 $\rightarrow$ 0.80	0.000	0.509	4.61
NGC5985	2.20	0.88	75.3	39.4	0.069	0.038	0.000	0.51 $\rightarrow$ 1.09	0.785	0.344	1.89
NGC0891	3.06	1.06	54.7	26.5	0.097	0.091	0.028	0.55 $\rightarrow$ 1.72	0.039	0.616	2.52
UGC02916	2.48	1.84	109.3	86.7	0.115	0.053	0.000	0.36 $\rightarrow$ 0.47	0.036	0.999	3.81
NGC4217	1.11	0.08	24.8	13.0	0.106	0.036	0.121	0.27 $\rightarrow$ 1.85	0.864	0.000	0.53
NGC4088	1.81	0.40	23.0	13.2	0.105	0.076	0.113	0.54 $\rightarrow$ 1.44	0.848	0.081	0.97
UGC05986	2.15	1.20	33.4	24.0	0.144	0.126	0.109	0.38 $\rightarrow$ 0.65	0.437	0.718	2.44
NGC3521	0.45	0.07	25.1	17.3	0.061	0.039	0.216	0.32 $\rightarrow$ 1.08	0.562	0.000	0.72
UGC12506	0.71	0.28	27.5	21.3	0.157	0.163	0.054	0.35 $\rightarrow$ 0.69	0.013	0.000	1.73

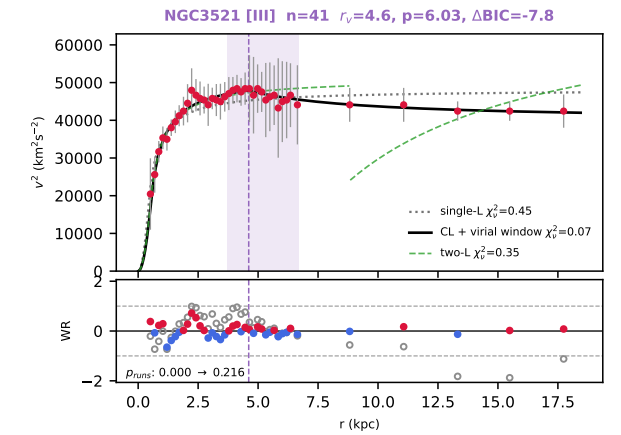
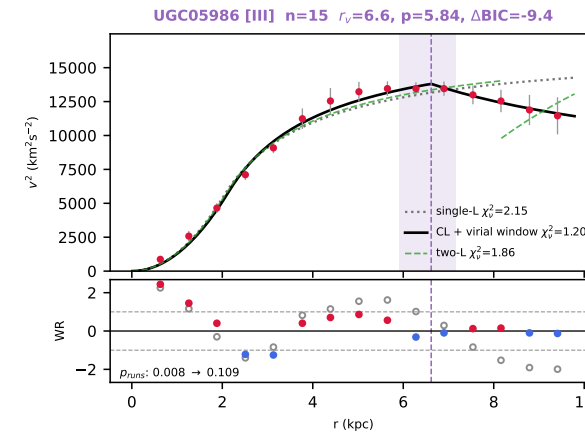
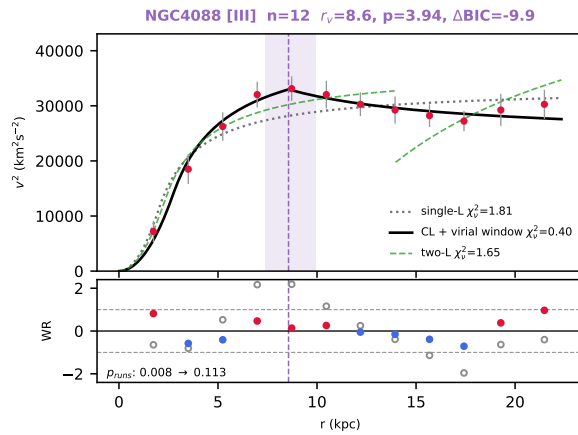
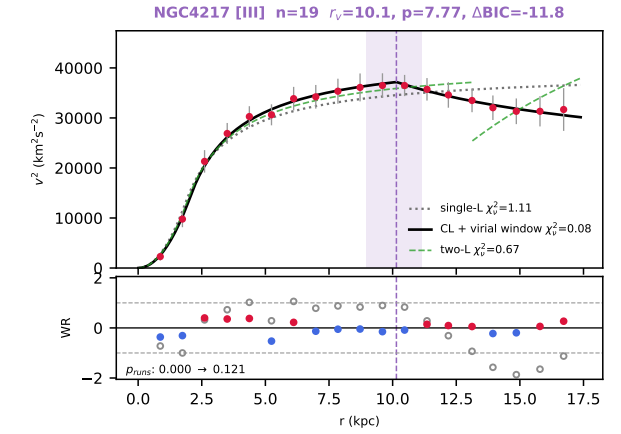
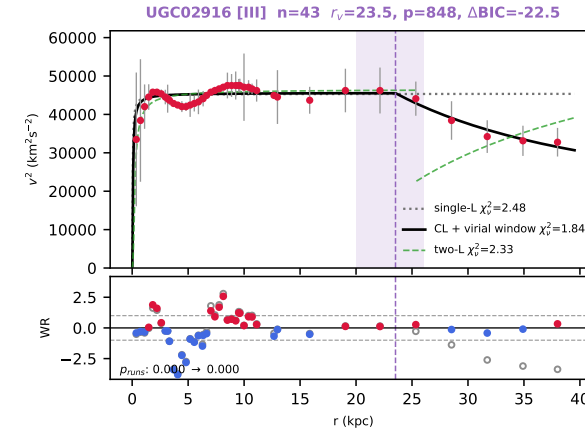
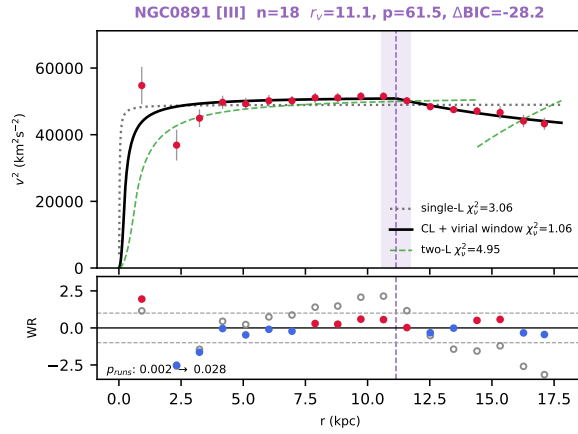
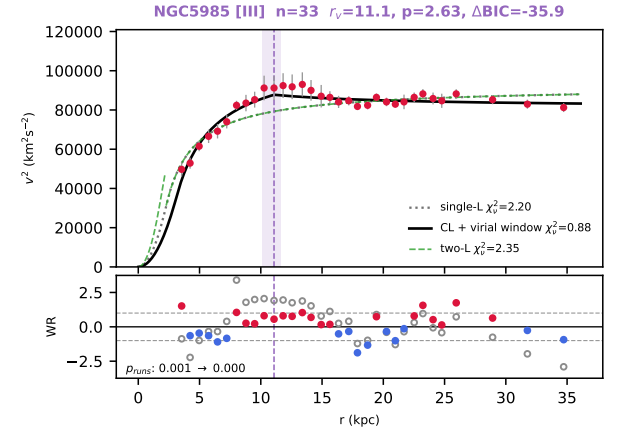
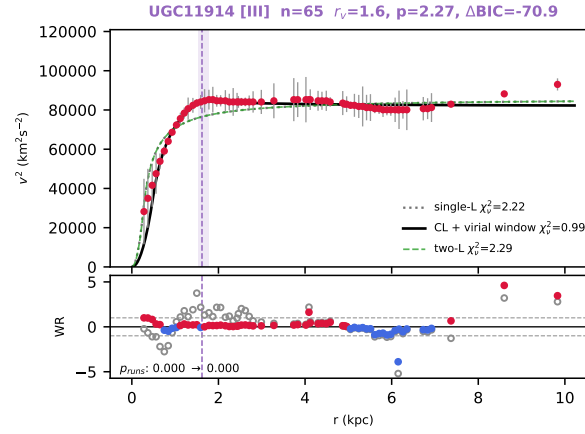
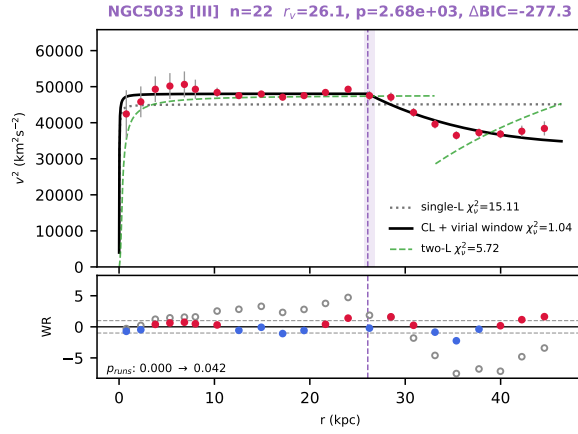
Galaxy	$\chi^2_{\nu,S}$	$\chi^2_{\nu,VW}$	BIC _S	BIC _{VW}	RMS _S	RMS _{VW}	$p_{\text{runs},VW}$	DW _S $\rightarrow$ DW _{VW}	p_{SW}	$P_{<}$	$\max z $
NGC5371	14.64	4.79	254.8	83.6	0.110	0.095	0.007	0.27 $\rightarrow$ 0.57	0.255	1.000	3.47
UGC11455	3.97	3.12	142.2	114.2	0.155	0.170	0.000	0.46 $\rightarrow$ 0.52	0.092	1.000	4.49
NGC6674	16.38	3.74	218.4	52.0	0.165	0.236	0.013	0.35 $\rightarrow$ 0.86	0.399	1.000	3.95
NGC5907	10.88	2.50	190.9	49.3	0.070	0.036	0.121	0.52 $\rightarrow$ 1.52	0.576	0.999	3.35
UGC02953	68.64	12.87	7765.5	1447.7	0.232	0.248	0.000	0.42 $\rightarrow$ 0.36	0.141	1.000	9.61
NGC5055	128.68	4.36	3352.5	117.9	0.313	0.120	0.004	1.07 $\rightarrow$ 1.23	0.031	1.000	5.80
NGC2903	26.86	2.10	866.5	77.1	0.355	0.100	0.006	0.18 $\rightarrow$ 0.84	0.351	1.000	3.02
ESO563-G021	3.38	0.94	101.5	38.2	0.116	0.089	0.071	1.03 $\rightarrow$ 2.34	0.002	0.456	3.07
IC4202	1.31	0.57	46.2	29.8	0.081	0.051	0.360	0.85 $\rightarrow$ 1.66	0.834	0.034	1.81
NGC6946	9.71	5.27	552.0	300.6	0.226	0.173	0.000	0.29 $\rightarrow$ 0.28	0.001	1.000	5.55

Atlas: category-C fits with residuals

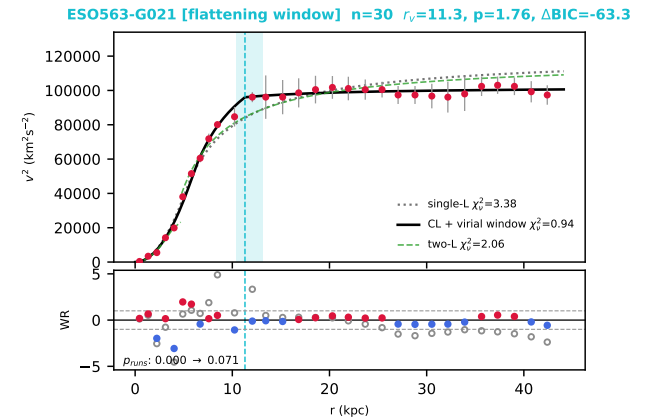
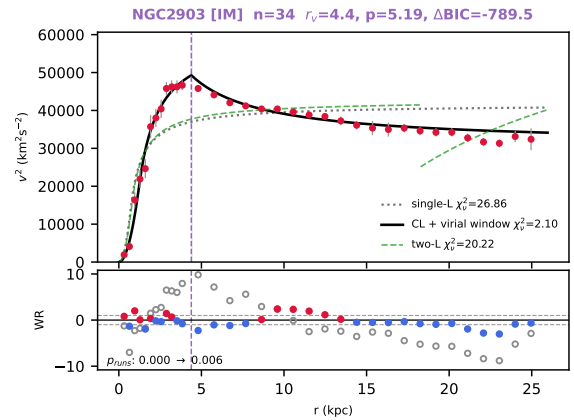
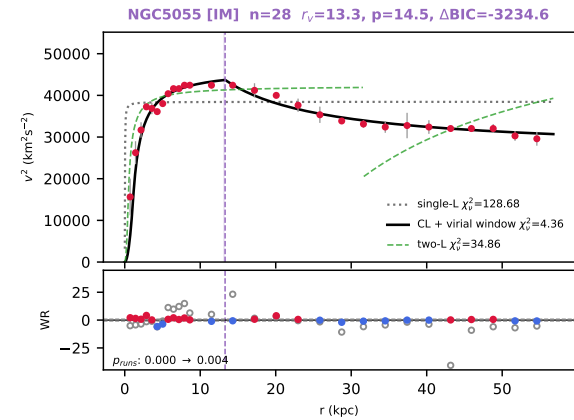
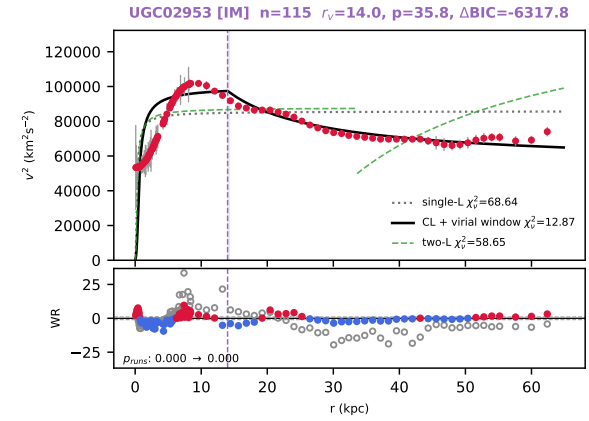
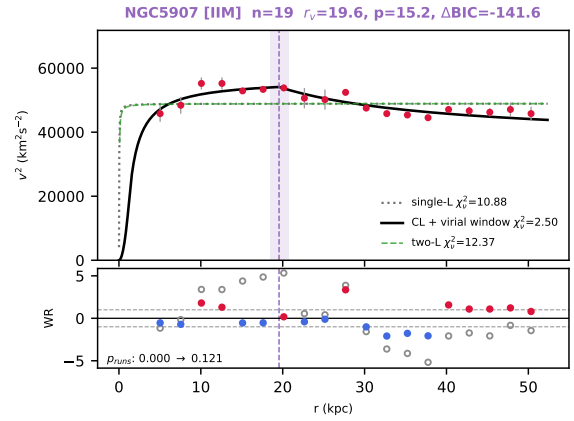
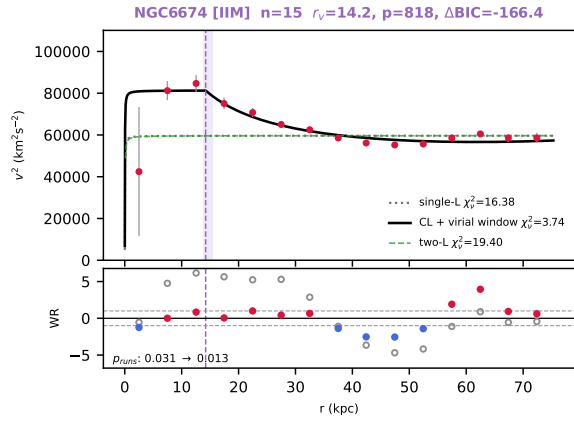
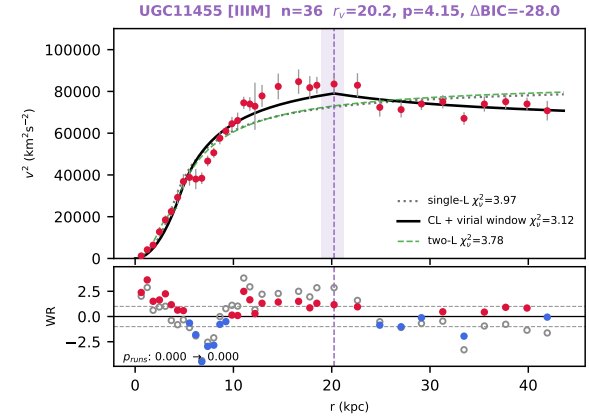
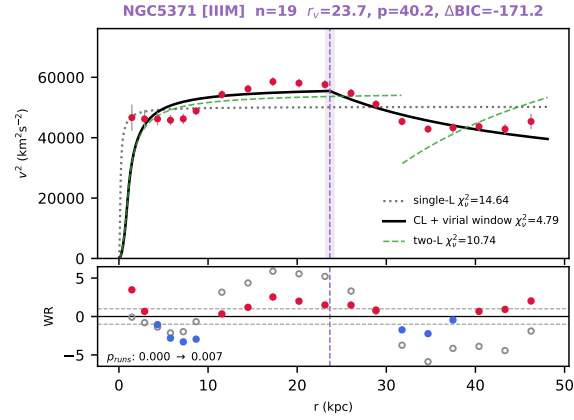
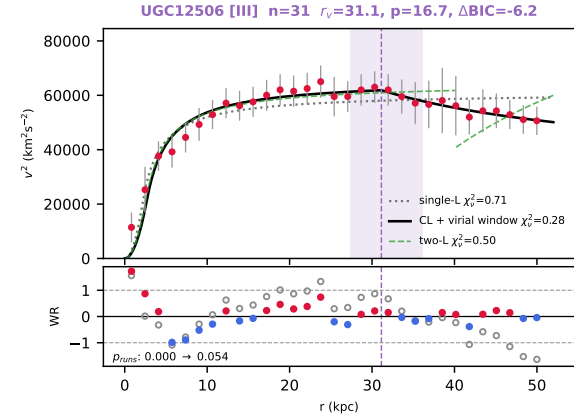
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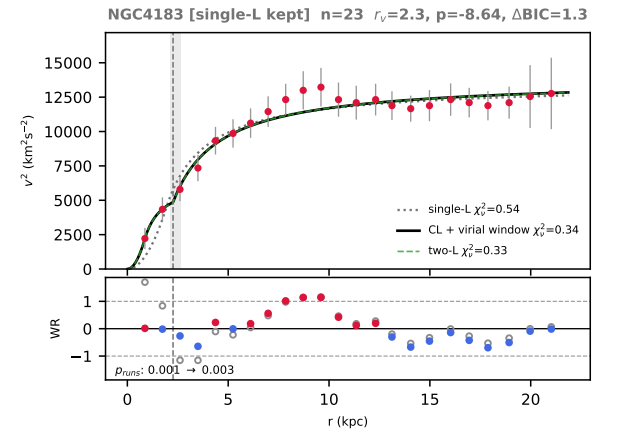
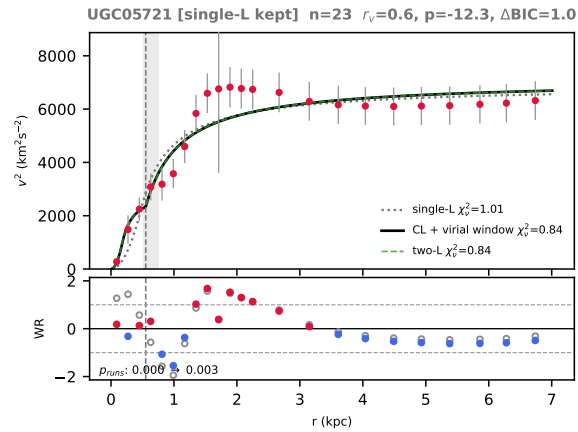
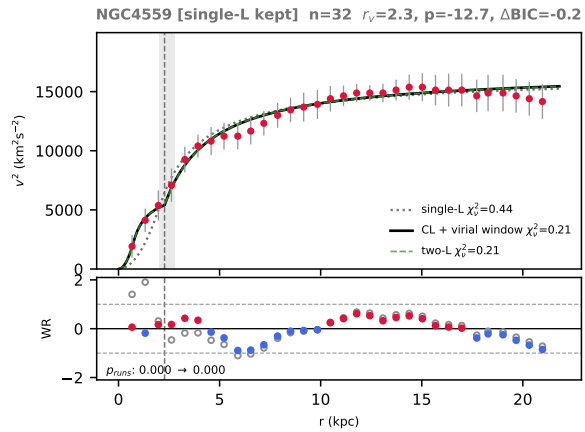
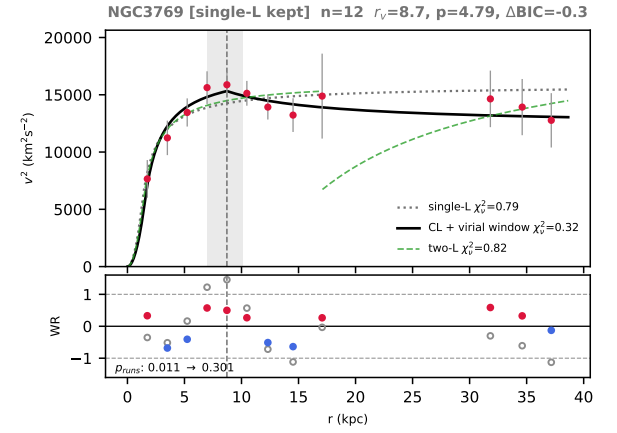
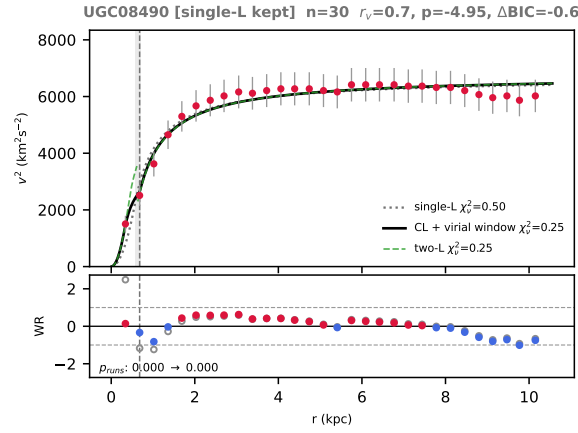
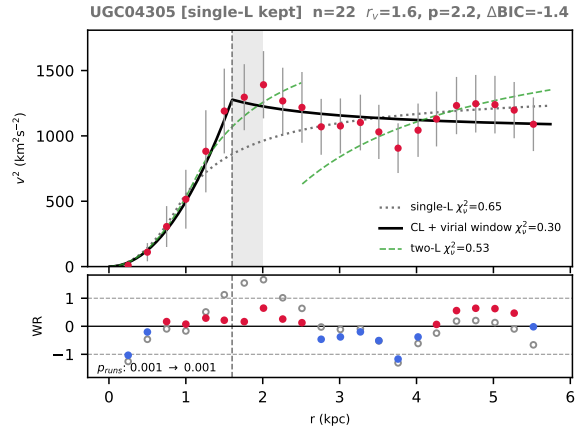
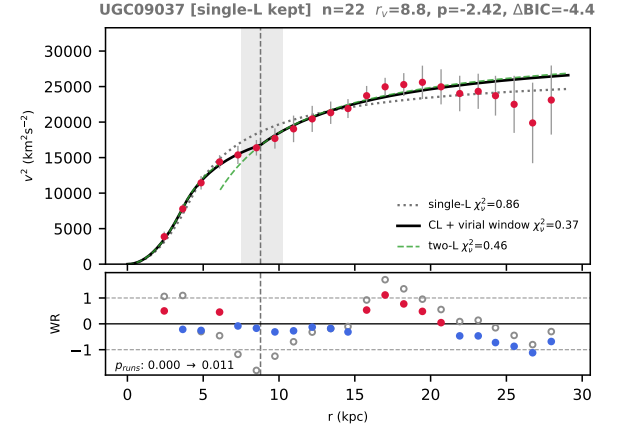
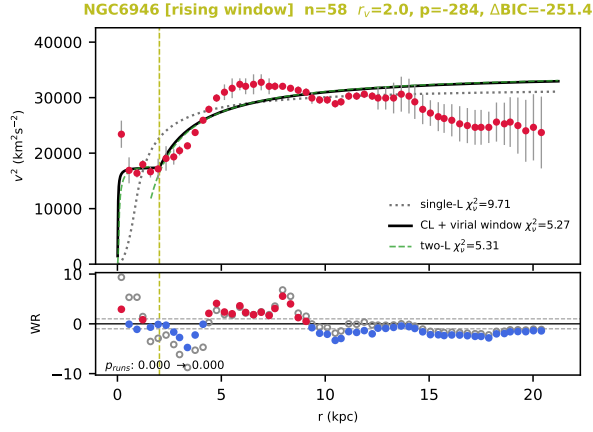
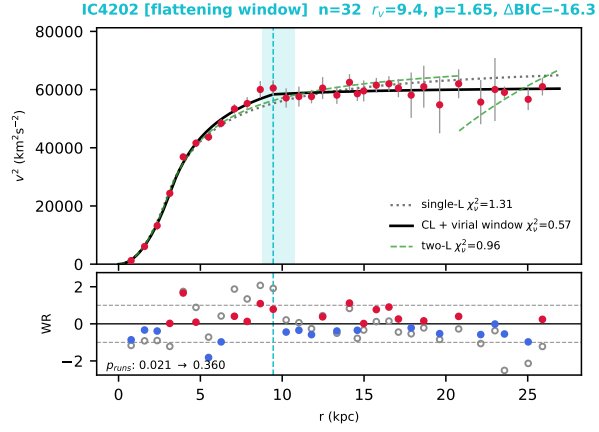
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Category C: single-L (dotted), CL + virial window (solid; dashed line r_v , band = profile $\Delta\chi^2 \leq 1$), two-L (green dashed). Residuals: single-L open, CL + VW filled.



Category C: single-L (dotted), CL + virial window (solid; dashed line r_v , band = profile $\Delta\chi^2 \leq 1$), two-L (green dashed). Residuals: single-L open, CL + VW filled.



Category C: single-L (dotted), CL + virial window (solid; dashed line r_v , band = profile $\Delta\chi^2 \leq 1$), two-L (green dashed). Residuals: single-L open, CL + VW filled.

